

Value-based Requirements Engineering for Human-centred AI: Foundations, Challenges, and a Research Agenda

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Abstract—Artificial intelligence (AI) systems are increasingly central to decisions that impact individuals and society, showing the need for these systems to align with stakeholder values. However, traditional engineering approaches often relegate values to implicit considerations or early design phases, which diminishes their influence on system behaviour and operational decision-making. This paper investigates how Value-based Requirements Engineering (VBRE) can facilitate the development of human-centred AI systems that uphold stakeholder values, especially within dynamic and collaborative human-AI contexts. We propose that VBRE serves as a foundational framework for integrating stakeholder values as first-class artefacts throughout the entire AI system lifecycle. Our analysis consolidates existing research and identifies three core challenges: (1) the multi-level operationalisation of values from abstract principles to measurable system behaviours, (2) closing the gap between design-time value specification and runtime system behaviour, and (3) enabling shared value-driven decision-making between humans and AI at runtime. This paper analyses these challenges and proposes a research agenda for advancing VBRE to support the development of value-aligned, transparent, and accountable human-centred AI systems.

Index Terms—Value-based Requirements Engineering; Human-centred AI; Stakeholder Values; Value Alignment; Shared Human-AI Decision-Making

I. INTRODUCTION

AI systems are increasingly involved in decisions that affect individuals and society, often operating in contexts where human values such as fairness, autonomy, and accountability are central. From healthcare diagnostics to criminal justice risk assessment, from autonomous vehicles to content moderation systems, AI systems make or recommend decisions that directly affect people’s rights, opportunities, and welfare. However, current engineering practices treat values as largely implicit, informal, or confined to early design stages [1], [2]. As a result, values are weakly integrated into the actual decision-making processes of AI systems, particularly in settings where decisions are shared between humans and AI [3]. While values shape expectations of system behaviour, they do not systematically guide how ethical and values-driven decisions are made, negotiated, or justified in operation. Existing approaches provide limited support for modelling

values as actionable constructs [1], reasoning about trade-offs [4], or enabling human–AI decision making grounded in those values [5]. This disconnection between abstract value statements and concrete system behaviour has been identified as a fundamental challenge in both software engineering and AI ethics literature [1], [6]. Furthermore, the rapid deployment of AI systems in high-stakes domains has outpaced the development of engineering methods that can ensure sustained value alignment throughout system operation [3].

The challenge is compounded by the inherent complexity of human values themselves: they are dynamic, pluralistic, and operate at multiple abstraction levels that frequently conflict [7], [8], none of which traditional requirements engineering was designed to handle.

We further argue that the current enthusiasm for technical AI alignment, through techniques such as Reinforcement Learning from Human Feedback (RLHF) [9] or Constitutional AI [10], does not substitute for values engineering at the level of requirements. The alignment challenge involves two interrelated parts: a *technical* part (how to encode values in systems) and a *normative* part (which values, for whom, and by what fair process) [6]. A system that is technically aligned with a narrow set of curated preferences may still violate the values of the communities it serves. Our central research question is: *How can stakeholder values be operationalised as traceable, verifiable, and adaptable requirements throughout the AI system lifecycle?*

In this paper, we argue that Value-based Requirements Engineering (VBRE) can play a central role in treating values as first-class artefacts that guide human-centred AI. We propose to build on VBRE to connect values to decision-making, support ethical reasoning, and enable shared human–AI decision processes. We further adopt a Values@Runtime perspective [11] whereby stakeholder values should not only be elicited and documented at design time, but represented as runtime-relevant concerns that can be monitored, reasoned about, negotiated, and revised as the AI system interacts with changing users, contexts, and social expectations. Specifically, we make the following contributions:

- We synthesise foundations from value-centred design, value-centred software engineering, AI ethics, and AI alignment to clarify the role of VBRE in engineering human-centred AI systems.
- We identify three core challenges for VBRE in AI: multi-level value operationalisation, maintaining alignment from design-time to runtime, and supporting shared value-driven decision-making between humans and AI.
- We position VBRE and technical AI alignment as complementary layers in a value-centred AI engineering stack, highlighting the need for explicit requirements work before values can be encoded in system behaviour.
- We outline a research agenda for advancing VBRE methods, tools, and educational practices to support traceable, verifiable, and adaptable value alignment throughout the AI system lifecycle.

The remainder of this paper is structured as follows. Section II establishes the foundations for understanding values in human-centred AI, drawing on socio-technical perspectives, values-to-requirements translation, value-centred design paradigms, and the limits of current technical AI alignment approaches. Section III identifies three central challenges for VBRE in AI systems: multi-level operationalisation, the gap between design-time specification and runtime behaviour, and shared value-driven decision-making between humans and AI. Section IV outlines a research agenda for advancing VBRE as a foundation for trustworthy and human-centred AI engineering. Finally, Section V concludes by calling for the requirements engineering community to treat values as first-class, traceable, and operational concerns throughout the AI system lifecycle.

II. FOUNDATIONS OF VALUES IN HUMAN-CENTRED AI

This section establishes the conceptual basis for treating values as first-class concerns in the engineering of human-centred AI systems. We first discuss values as socio-technical constructs that are dynamic, context-dependent, and shaped through interaction between people, organisations, and technologies. We then examine how values can be translated into requirements, before reviewing existing value-centred paradigms and clarifying how they relate to, and differ from, current approaches to technical AI alignment.

A. Values as Socio-technical Constructs

Values are guiding principles that influence decisions and behaviour across individuals, organisations, and societies. In the seminal theory of basic human values, Schwartz identifies ten universal value types (e.g. self-direction, security, benevolence, and universalism) that are organised within a circular motivational continuum [7]. In this structure, adjacent values are mutually compatible, whereas opposing values generate tension. This psychological foundation offers a structured framework for understanding values as enduring motivational goals that transcend specific situations and guide the selection and evaluation of behaviour and events.

However, values cannot be understood solely as individual psychological constructs. They are fundamentally sociotechnical phenomena that both shape and are shaped by interactions among people, technologies, and broader social contexts. Friedman *et al.*'s [12] Value Sensitive Design (VSD) framework emphasises that values are embedded in technological systems through design choices, while these systems simultaneously influence human behaviour and social practices. This bidirectional relationship implies that technological systems, including AI systems, not only reflect the values of their designers and stakeholders but also actively participate in shaping the values that emerge within their contexts of use.

Values operate across multiple levels of abstraction, ranging from high-level ethical principles to context-specific interpretations and, ultimately, to concrete behavioural instantiations [8], [12]. At the most abstract level, values such as fairness or autonomy function as broad normative commitments that often attract widespread agreement. Yet, these abstract principles must be interpreted and specified within particular domains, stakeholder groups, and use contexts [8], [13]. For example, fairness in a hiring AI system may be operationalised differently than fairness in a medical diagnosis system, reflecting distinct stakeholder priorities, regulatory requirements, and technical constraints [1]. At the most concrete level, values must be translated into system behaviours, interface designs, and decision-making procedures that embody these commitments in practice [8], [12].

Importantly, values are inherently subjective, context-dependent, and dynamic [7], [12]. Different individuals and communities prioritise values in diverse ways, giving rise to value pluralism and, at times, conflict [7], [14]. Furthermore, values evolve over time in response to shifting social norms, technological developments, and lived experiences [12]. This dynamic and pluralistic nature presents significant challenges for engineering systems that must remain aligned with stakeholder values throughout their lifecycle [1]. Consequently, a sociotechnical perspective calls for AI systems to be designed not as static implementations of fixed value specifications, but as adaptive systems capable of ongoing negotiation, reflection, and alignment with evolving human values [12].

B. From Values to Requirements

In Requirements Engineering, values have traditionally been treated as high-level concerns that influence goals, constraints, and trade-offs, often captured as softgoals or non-functional requirements [13]. Thew and Sutcliffe's [13] value-based requirements engineering method provides structured techniques for eliciting stakeholder values, linking them to system goals, and negotiating value conflicts during requirements specification. Their approach recognizes that values provide the "why" behind functional requirements, explaining the deeper motivations and priorities that drive system design.

However, values extend beyond specifying what a system should do, they also shape how decisions are made and who is involved in making them. This is particularly critical in AI systems where decision authority may be distributed between

humans and automated components, and where the basis for decisions must be transparent and accountable. Van de Poel [8] argues that translating values into design requirements involves a process of specification, whereby abstract values are progressively refined into concrete design criteria through a series of interpretive and technical choices. This translation process is not purely technical but involves normative judgments about how values should be balanced, prioritized, and instantiated in specific contexts.

Recent work has emphasized the need for values traceability throughout the software lifecycle, enabling stakeholders to understand how high-level value commitments are reflected in system requirements, design decisions, and implemented behaviours [1], [15]. Perera's [15] goal modelling approach provides mechanisms for linking values to goals and requirements, creating explicit traceability chains that support verification and validation of value alignment. Such traceability is essential for ensuring that values are not lost or diluted as they are translated through successive stages of system development.

The relationship between values and requirements is further complicated by the need to support value-driven decision-making at runtime [11], [16]. While traditional requirements engineering focuses on specifying system behaviour at design time [13], AI systems often operate in dynamic environments where decisions must be made in response to unforeseen situations and evolving stakeholder needs [3], [11]. This demands a shift from viewing requirements as static specifications to treating them as dynamic constraints and guidance mechanisms that support ongoing value alignment [11]. Requirements must not only specify what the system should do but also provide the basis for ethical reasoning [16], [17], trade-off analysis [4], and human-AI collaboration during operation [5], [18].

C. Value-centred Paradigms

Several established paradigms provide important foundations for integrating values into software and AI engineering. Value Sensitive Design (VSD), introduced by Friedman *et al.* [12], is the canonical framework for systematically accounting for human values in technology design. VSD employs a tripartite methodology consisting of conceptual investigations (identifying relevant stakeholders and values), empirical investigations (studying how stakeholders and technologies interact), and technical investigations (designing systems that support identified values). This iterative approach has been widely adopted in human-computer interaction and information systems research, demonstrating the feasibility of value-centred design processes.

However, a systematic review by Gerdes and Frandsen [2] reveals that VSD research has been dominated by conceptual and empirical work, with relatively limited attention to technical investigations and verification methods. This gap is particularly problematic for AI systems, where the complexity of machine learning models and the opacity of decision-making processes make it difficult to verify that implemented

systems actually embody intended values. The review calls for more rigorous technical methods to operationalise, implement, and verify value alignment in software systems.

Value-based Requirements Engineering (VBRE) provides complementary methods specifically focused on the requirements phase of system development. Thew and Sutcliffe's [13] VBRE approach includes techniques for value elicitation through stakeholder workshops, value modelling using goal-oriented notations, and value negotiation through structured dialogue. Harbers *et al.* [19] extend this work by proposing methods for embedding stakeholder values throughout the RE process, including participatory modelling techniques that engage stakeholders in collaborative value specification. Detweiler and Harbers [20] introduce "value stories" as a narrative method for discussing and identifying stakeholder values, providing a more accessible alternative to formal modelling notations.

Recent research has developed tools and frameworks for operationalising values in software engineering practice. Ferrario and Winter [21] adapt Schwartz's theory of basic values to software engineering contexts, providing empirical methods such as the Values Q-Sort for measuring value priorities among software engineers and stakeholders. Nurwidiantoro *et al.* [4] propose a "human values dashboard" that visualises value considerations and trade-offs throughout software development, supporting ongoing awareness and monitoring of value alignment. Shahin *et al.* [1] provide a comprehensive survey of 51 studies on operationalising human values in software engineering, identifying common approaches, challenges, and research gaps.

Despite these advances, existing paradigms primarily focus on early lifecycle stages (i.e., requirements elicitation, conceptual design, and initial implementation) and provide limited support for runtime behaviour, human-AI interaction, and ethical reasoning in decision-making [2]. The challenge of maintaining value alignment throughout system operation, particularly in dynamic and uncertain environments, remains largely unaddressed [11]. Furthermore, most existing approaches were developed for traditional software systems and have not been fully adapted to the unique challenges of AI systems, including learning from data, adapting behaviour over time, and operating with limited human oversight [2], [3]. To bridge this gap we may need to extend VBRE paradigms to explicitly address the runtime dimension of value alignment and the specific characteristics of AI technologies.

D. AI Alignment

A growing body of work in AI addresses the problem of alignment to ensure that AI system behaviour corresponds to human intentions and values. Gabriel [6] argues that alignment must ultimately be grounded in values themselves, and this challenge is as much normative and political as it is technical. Because people reasonably disagree about values, the goal is not to encode a single correct set of values but to establish fair, inclusive processes for deciding which values AI systems

should uphold, which is a task that belongs also to requirements engineering, not just to model training.

Current technical alignment approaches, notably Reinforcement Learning from Human Feedback (RLHF) [9], Constitutional AI [10], and reward modelling [22] operate at the level of model behaviour optimisation. They presuppose a set of preferences or value assumptions and tune models to approximate them. They do not, in themselves, address: (a) how diverse stakeholder values are elicited and represented before they can inform training signals; (b) how the operationalisation of values is traced through system design decisions; (c) how value conflicts among different stakeholder groups are negotiated; or (d) how compliance with stated value commitments is verified and governed at runtime.

Calinescu *et al.* [16] make a complementary argument from a formal methods perspective. Their SLEEC norm operationalisation process demonstrates that translating normative principles into verifiable AI agent behaviour requires a systematic engineering process of multiple stages: 1) capability specification, 2) requirements elicitation, 3) well-formedness checking, 4) SLEEC-aware implementation, and 5) compliance verification. They show that this process can and should surface infeasible value combinations before deployment, rather than discovering violations in operation. This is precisely the kind of structured requirements engineering process that technical alignment methods presuppose but do not provide.

Technical alignment thus starts where VBRE stops. A well-executed VBRE process would produce the value commitments, priority orderings, and stakeholder agreements that technical alignment methods need as input. Without this upstream engineering work, the values encoded in alignment procedures reflect the implicit choices of model developers rather than the explicit, negotiated commitments of affected communities. We position VBRE and technical alignment as *complementary layers* in a value-centred AI engineering stack: VBRE provides the sociotechnical requirements layer (what values, for whom, at what priority); technical alignment implements the model-level behavioural layer (how the system approximates those values in practice). Neither layer is sufficient without the other, and the current imbalance toward the latter at the expense of the former is a structural problem in AI engineering practice.

III. CHALLENGES

This section identifies three challenges that must be addressed for VBRE to support human-centred AI in practice. These challenges arise from the difficulty of translating abstract and contested values into actionable system requirements, maintaining value alignment as AI systems operate in dynamic environments, and supporting decision-making processes in which authority and accountability are shared between humans and AI. Together, they show why value alignment cannot be treated as a one-off design activity, but must be engineered, monitored, and adapted across the AI system lifecycle.

A. Multi-level Operationalisation

Operationalising values requires bridging multiple levels of abstraction: from abstract ethical concepts, to context specific interpretations, to measurable behavioural instantiations that can guide system implementation and operation. This multi-level translation process is one of the most fundamental challenges in values-based requirements engineering, as each level involves different stakeholders, languages, and verification criteria [8].

At the highest level, values are expressed as abstract principles such as "fairness", "privacy", "autonomy", or "accountability". These principles command broad agreement but provide limited guidance for system design without further specification. The first operationalisation challenge is to interpret these abstract values in the context of specific application domains, stakeholder groups, and use cases. For instance, the value of "fairness" in an AI hiring system might be interpreted as equal opportunity, demographic parity, or individual fairness, each with different implications for system design and evaluation [1]. This interpretive step requires engaging diverse stakeholders to understand their value priorities and negotiating shared understandings of what values mean in context [13], [19].

The second level of operationalisation involves translating context-specific value interpretations into design requirements and technical specifications. Van de Poel [8] describes this as a process of "specification", whereby abstract values are progressively refined through a series of design decisions that constrain the solution space. For example, a commitment to privacy might be specified as requirements for data minimisation, user consent mechanisms, and secure data storage. However, this translation is not straightforward, as multiple technical approaches may be consistent with a given value interpretation, and different approaches may have different implications for other values. Kapferer *et al.* [23] demonstrate how domain-driven design practices can be adapted to support value-driven analysis and design, providing structured methods for linking values to architectural decisions.

The third level of operationalisation involves implementing and verifying that system behaviour actually instantiates the intended values. This requires measurable indicators and verification methods that can assess whether the system behaves in value-aligned ways. Perera's [15] work on values traceability provides goal modelling techniques that link values to requirements to implementation artefacts, enabling verification that value commitments are preserved throughout development. Ferrario and Winter [21] propose empirical methods for measuring human values in software engineering contexts, including the Values Q-Sort tool that can assess value priorities among developers and stakeholders. However, measuring value alignment in AI systems is particularly challenging due to the complexity of learned behaviours and the difficulty of anticipating all possible operational scenarios.

As illustrated in Figure 1, each transition between abstraction levels is a potential site of *abstraction drift*, where lower

levels lose fidelity to higher level value commitments.

A key challenge in multi-level operationalisation is maintaining coherence and traceability across these abstraction levels. As values are translated from abstract principles to concrete implementations, there is a risk of drift, where the implemented system no longer reflects the original value commitments. We refer to this phenomenon as *value drift*: the progressive divergence between the values elicited and operationalised at design time and the values enacted by the system during use. Value drift may occur because stakeholder priorities change, because the operational environment shifts, or because learned system behaviour produces effects that were not anticipated during requirements engineering [3]. Shahin *et al.*'s [1] survey identifies the lack of systematic methods for operationalising values across abstraction levels as a critical gap in current software engineering practice. Furthermore, different stakeholders may operate at different levels of abstraction, with end-users focused on concrete behaviours, developers on technical specifications, and ethicists on abstract principles. Bridging these perspectives requires methods that support multi-level reasoning and communication about values.

B. From Design-Time to Runtime: Closing the Gap

Values elicited at design-time often fail to capture how systems behave in practice, creating a critical gap between intended and actual value alignment. This gap arises from several sources: values may evolve over time as social norms and stakeholder priorities change, operational contexts may differ from those anticipated during design, and the complexity of AI systems may lead to emergent behaviours that were not foreseen by designers. Existing requirements engineering approaches, which focus primarily on design-time specification, lack mechanisms for runtime monitoring, feedback, and adaptation of value alignment [11].

The design-time to runtime gap is particularly acute for AI systems that learn from data and adapt their behaviour over time. Machine learning models may exhibit biases or unintended behaviours that only become apparent when deployed in real-world contexts [3]. For instance, a hiring AI trained on historical data may perpetuate past discrimination patterns, violating fairness values despite careful design-time specification [3]. Moreover, AI systems often operate in dynamic environments where the distribution of inputs, the preferences of users, and the broader social context may shift over time, requiring ongoing adaptation to maintain value alignment.

A runtime perspective on values enables them to be represented, monitored, analysed, and aligned during system operation. Nurwidyantoro *et al.*'s [4] human values dashboard provides visualisation and monitoring tools that track value considerations throughout software development and operation, supporting ongoing awareness of value trade-offs and potential misalignments. However, such monitoring requires operationalised value indicators that can be measured at runtime, as well as mechanisms for detecting when system behaviour diverges from value commitments. Perera *et al.*

[24] demonstrate empirically that considering human values during requirements engineering activities leads to different design decisions and priorities, suggesting that runtime value monitoring could similarly influence operational decisions.

Closing the design-time to runtime gap requires mechanisms for value adaptation and realignment. When mismatches between stated values and observed behaviour are detected, systems must be able to adapt their behaviour or engage human stakeholders in decision-making. Tubella *et al.*'s [17] concept of governance by glass-box proposes transparent mechanisms for implementing moral bounds on AI behaviour, enabling runtime monitoring and intervention when values are at risk. Rodriguez-Soto *et al.* [25] demonstrate how multi-objective reinforcement learning can be used to instill moral value alignment in multi-agent systems, providing algorithmic approaches for balancing competing values during operation. However, these technical approaches must be integrated with requirements engineering processes that specify when and how adaptation should occur.

The challenge of maintaining value alignment at runtime also raises questions about the role of human oversight and intervention. While some value violations may be addressed through automated adaptation mechanisms, others may require human judgment and decision-making. Requirements must specify the conditions under which human intervention is required, the information that should be provided to support human decision-making, and the mechanisms for incorporating human feedback into system behaviour. This demands a shift from viewing AI systems as autonomous agents to treating them as collaborative partners in value-driven decision making [5], [26].

C. Shared Decision-Making at Runtime

AI systems increasingly operate in collaboration with humans rather than as standalone decision makers, raising fundamental questions about how decision authority should be distributed and how values should guide human-AI interaction. Shared decision-making at runtime involves determining when the system should decide autonomously, when humans should intervene, and how decisions should be negotiated when values are at stake. Requirements engineering must explicitly capture and support these shared decision-making processes, with values providing the basis for structuring interaction and guiding trade-offs.

The challenge of shared decision-making is rooted in the recognition that AI systems and humans have complementary strengths and limitations. AI systems can process large amounts of data, identify patterns, and make consistent decisions at scale, but they lack contextual understanding, common sense reasoning, and the ability to make nuanced ethical judgments [3], [6]. Humans bring contextual knowledge, ethical reasoning capabilities, and accountability, but are subject to cognitive biases, limited attention, and inconsistent decision-making. Effective human-AI collaboration requires allocating decision authority in ways that leverage the strengths of both while mitigating their limitations.

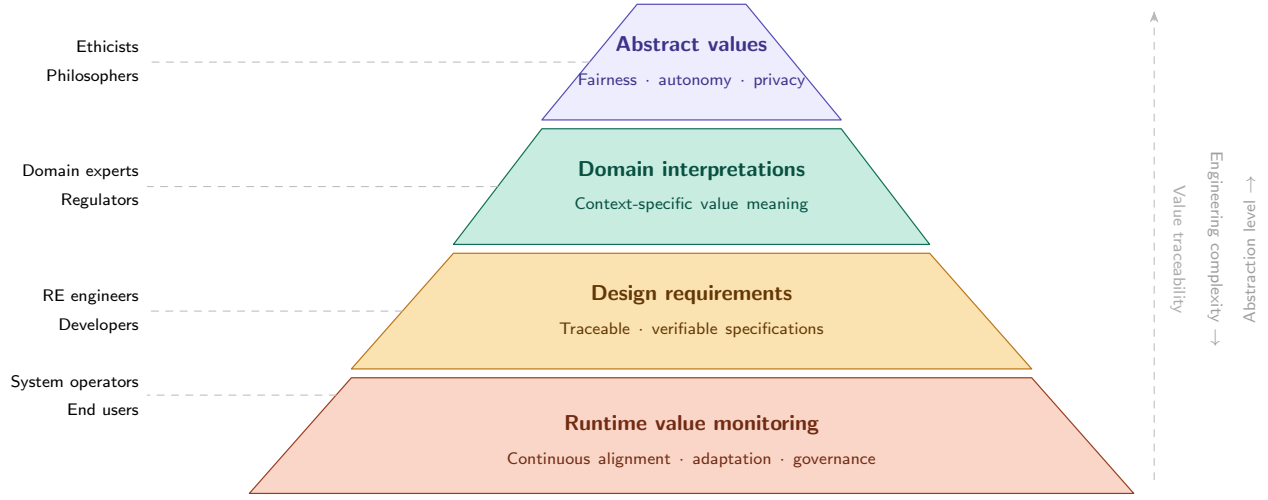


Fig. 1. Multi-level value operationalisation pyramid. Abstraction drift occurs when lower levels lose fidelity to higher-level value commitments; each band transition is therefore a verification checkpoint.

Values play a central role in determining appropriate decision authority allocation. For instance, decisions that significantly impact individual autonomy or dignity may require human involvement, even if an AI system could make technically accurate predictions. Conversely, decisions that require consistency and freedom from human bias may be better delegated to AI systems, with human oversight to ensure value alignment. Aizenberg and Van Den Hoven [26] argue that designing for human rights in AI requires explicit consideration of which decisions should remain under human control and which can be delegated to automated systems. This allocation of decision authority is itself a value-laden choice that must be negotiated with stakeholders during requirements engineering [13], [18].

Requirements for shared decision-making must specify the conditions under which human intervention is triggered, the information provided to support human judgment, and the mechanisms for incorporating human decisions into system behaviour. McIntosh *et al.* [5] demonstrate how the ACT Matrix, a tool from acceptance and commitment therapy, can be adapted to elicit human values-driven user requirements that explicitly address human-AI interaction patterns. Their case study shows how value-based requirements can guide the design of interfaces and interaction protocols that support meaningful human involvement in AI-mediated decisions. However, designing effective human-AI collaboration requires understanding not only what values are at stake but also how humans and AI systems can communicate about values and coordinate their decision-making.

Bergman *et al.*'s [18] STELA framework provides a community-centred approach to norm elicitation for AI alignment, emphasising participatory methods that engage diverse stakeholders in specifying how AI systems should behave in value-laden situations. This participatory approach recognises that shared decision-making is not merely a technical problem

but a social and political one, requiring ongoing dialogue and negotiation among stakeholders with different value priorities. Sorensen *et al.*'s [14] roadmap to pluralistic alignment further emphasises the need for AI systems that can accommodate diverse and sometimes conflicting values, rather than imposing a single value framework. Requirements engineering for shared decision-making must therefore support value pluralism and provide mechanisms for negotiating value conflicts at runtime.

IV. RESEARCH AGENDA

We conclude with a research agenda for VBRE in human-centred AI. These directions represent structural gaps that current paradigms do not address. We argue that closing them is a prerequisite for responsible AI engineering at scale.

Integrating VBRE with norm operationalisation frameworks. The SLEEC norm operationalisation process [16] demonstrates what rigorous, verifiable translation of normative principles into AI agent requirements can look like. We call for systematic work to integrate such frameworks into VBRE methodologies, so that the formal checking, conflict detection, and compliance verification that SLEEC provides at the agent level become standard parts of RE for AI more broadly. This integration must also address the challenge of stakeholder engagement across technical and non-technical roles [18], [19].

Participatory and story-based value elicitation. Static, workshop-based elicitation methods [13] are insufficient for capturing values that are dynamic, contested, and distributed across diverse stakeholder communities [11]. Future methods must adopt participatory frameworks that engage stakeholders as ongoing co-designers of value commitments, and story-based approaches [20] that accommodate the temporal fluidity of values. This is especially important considering Gabriel's

argument [6] that fair alignment principles must be identified through inclusive processes rather than imposed by developers.

Verifiable, multi-level operationalisation frameworks.

There is a need for formal or semi-formal frameworks that trace value commitments from stakeholder intent through domain-specific interpretation to measurable technical indicators, preserving fidelity at each level. Such frameworks must support verification that system behaviour at runtime is actually consistent with upstream value commitments, not just at deployment but throughout operation. Sorensen *et al.*'s [27] Value Kaleidoscope offers a promising model for how pluralistic values can be represented and reasoned about computationally, linking such representations to requirements artefacts is an important open problem.

Values@Runtime and adaptive re-alignment.

VBRE must extend its temporal scope to encompass the full operational lifecycle of AI systems. A Values@Runtime perspective [11] requires mechanisms for representing values as runtime-relevant artefacts, linking them to observable indicators, detecting value drift, and triggering appropriate human or automated responses. This includes not only technical monitoring, but also governance processes through which stakeholders can contest, reinterpret, or revise value commitments as operational experience accumulates. The hard/soft ethics framework described by Inverardi [28] provides a useful lens: hard value commitments (mandatory, verified) and soft value commitments (configurable, user-adjustable) may require different monitoring and adaptation mechanisms, and requirements engineering must provide the architecture for both.

Multidisciplinary education for value-centred AI engineering. As Calinescu *et al.* [16] note, SLEEC-aware engineering requires competence that cuts across ethics, law, domain expertise, and software engineering, and most practitioners currently possess only a subset. We call on the RE and software engineering communities to develop curricular frameworks and shared educational resources that prepare multidisciplinary teams to co-create AI systems that are not only technically capable but demonstrably aligned with the values of the communities they serve.

V. CONCLUSION

This paper has argued for a fundamental reorientation of how the AI engineering community thinks about values. Value-based Requirements Engineering is not a supplementary concern to be addressed after the core system is built, it is the foundational discipline that makes trustworthy human-centred AI possible. Technical alignment methods, however sophisticated, presuppose the upstream requirements work that VBRE provides: the elicitation of diverse stakeholder values, the operationalisation of those values into traceable and verifiable requirements, and the governance mechanisms that maintain alignment throughout system operation. The three challenges we have identified : 1) multi-level operationalisation, 2) closing the design-time to runtime gap, and

3) enabling shared value-driven decision-making. Addressing these challenges requires new methods, new tools, and multidisciplinary collaboration. Without progress on these fronts, value alignment risks becoming a theoretical claim rather than a verifiable property, one whose failures are most acutely felt by those with the least power to contest them. We call on the requirements engineering community to take ownership of these problems and to establish VBRE as a first-class discipline for the AI era.

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